⚠ IMPORTANT — Support Scope

3DPTronics supports the VirtuaTilt controller hardware only. This guide is a *compendium* to help you connect your VirtuaTilt to third-party pinball software (VPX, Future Pinball + BAM, DOF/DOFLinx, PinEvent, PinUP Popper, etc.). We provide them as a courtesy to save you the hunt across multiple sites. **The information is best-effort** and kept reasonably current, but is **not** official documentation of those projects. **We do not provide official support for any third-party software referenced here** — for software-specific issues, refer to each project's own docs and community channels.

☑ Officially supported

Component	Coverage
🎮 **VirtuaTilt controller**	Wired and Wireless editions, all hardware components
💾 **Pinscape Pico firmware**	Updates, OTA flashing, recovery via Updater Tool
📋 **Bundled profiles**	All BT and USB profiles shipped in the redist
⚙ **GUIConfigTool**	Configuration, calibration, profile loading, output testing
🔧 **Wireless Kit**	Wireless hardware kit for existing VirtuaTilt owners
📦 **Accessories**	Anything shipped directly by 3DPTronics

❌ Not supported

Category	Examples	Where to get help
🎲 **VP Game engines**	VPX, Visual Pinball, Future Pinball, BAM	VPForums · VP Universe
🕹 **VP titles**	Pinball FX / FX2 / FX3 / VR / M, etc.	Zen Studios support
💥 **Feedback frameworks**	DOF (Direct Output Framework), DOFLinx	DOF on GitHub · DOFLinx wiki
🎨 **DMD / extras**	DMDExt, PinEvent V2, PuP-Packs / PUP-Stream	VP Universe
🚀 **Frontends**	PinUP Popper, Baller Installer, etc.	Popper forum
🎬 **Emulators / ROMs**	MAME, pinball ROMs	MAMEdev
💻 **System-level**	Windows, GPU/audio drivers, USB controllers	Microsoft / vendor support

🌐 Community help for software

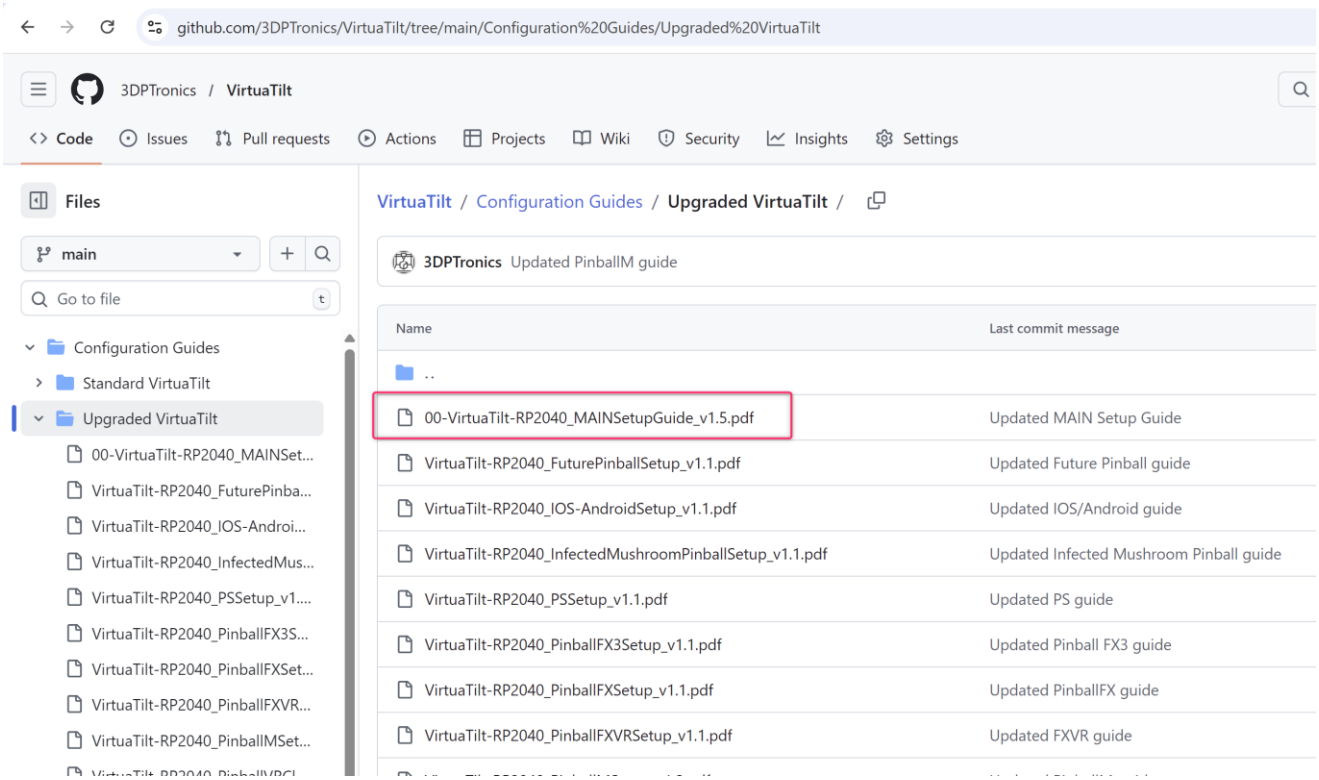
Project	Community
Visual Pinball X	vpforums.org · vpuniverse.com
Future Pinball + BAM	VPU Future Pinball forum
Future Pinball + BAM AIO	VPU file 14807 — FP+BAM Essentials AIO
DOF	github.com/DirectOutput/DirectOutput
DOFLinx	doflinx.github.io/docs
Pinball FX family	Zen Studios Support
PinUP Popper	VPU PinUP Popper forum
MAME	mamedev.org

💡 **Why this matters:** Our team is small and our expertise is the VirtuaTilt hardware — we can't debug VPX scripts, flaky BAM installs, or broken DOF chains. Those communities have their own experts. The **AI chatbot on the wiki** can help connect VirtuaTilt with most of the software above; for deeper or vendor-specific issues, reach out to the appropriate community first.

VirtuaTilt – Firmware Update Instructions

To update firmware with official Pinscape Pico releases (https://github.com/mjrgh/PinscapePico/releases) or custom firmware releases (available in this private share folder) please refer to **MAIN SETUP Guide** on our GitHub repository or have a look at Device Setup video tutorial on our YouTube channel:

https://github.com/3DPTronics/VirtuaTilt/tree/main/Configuration%20Guides/Wired%20VirtuaTilt



https://www.youtube.com/watch?v=bB6r3Rt6KcY



VirtuaTilt Tutorials - Device Setup

Following some basic instructions to have your new VirtuaTilt up and running with Pinball M and its new DOFLinx support (full feedback!).

Download **DOFLinx** latest package from here: Releases · DOFLinx/DOFLinx (github.com)

Download also **B2S backglass set** for DOFLinx with Pinball M: Release B2S Back glass set for DOFLinx with Pinball M V2 · DOFLinx/B2S-Back-Glasses · GitHub

NOTE: Latest DOFLinx is required for Pinball M.

Please always refer to official DOFLinx guides available at:

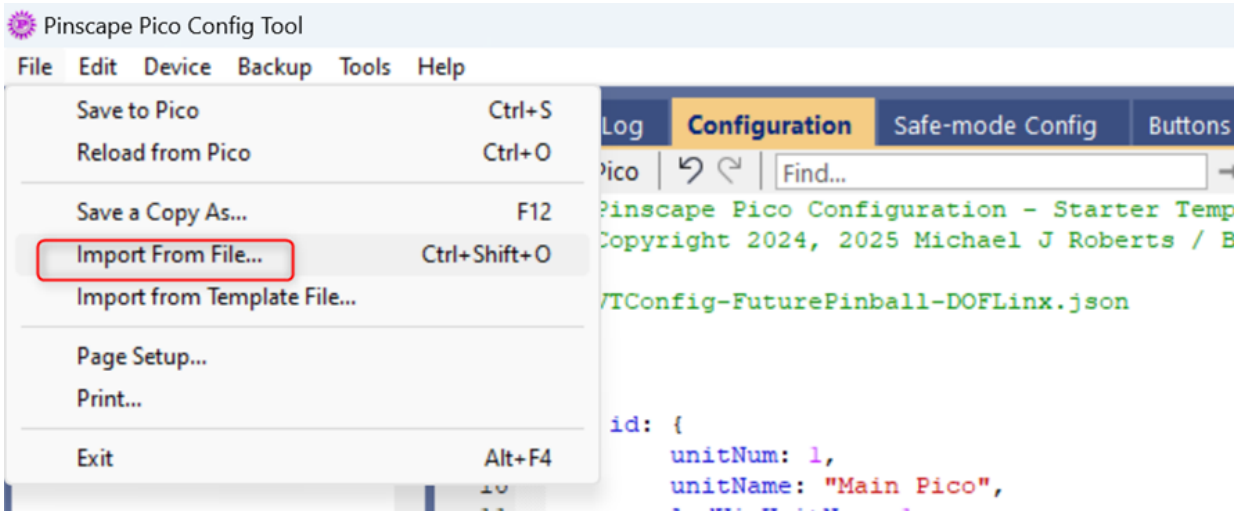
https://doflinx.github.io/docs/

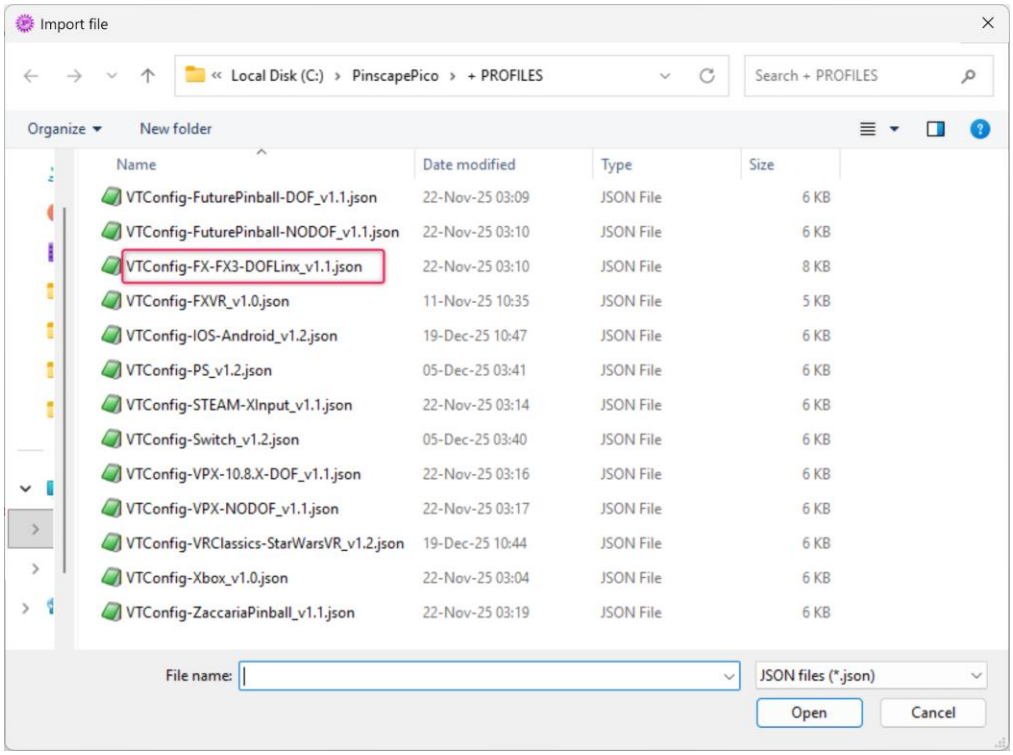
https://doflinx.github.io/docs/getting-started/08_PinballM.html

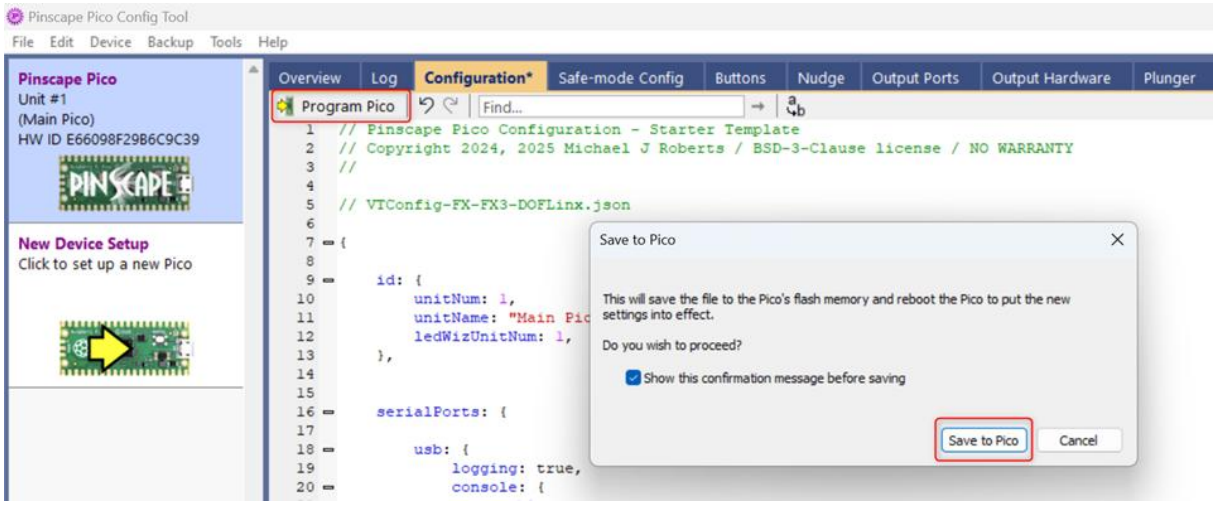
DOFLinx Setup

PREREQUISITES:

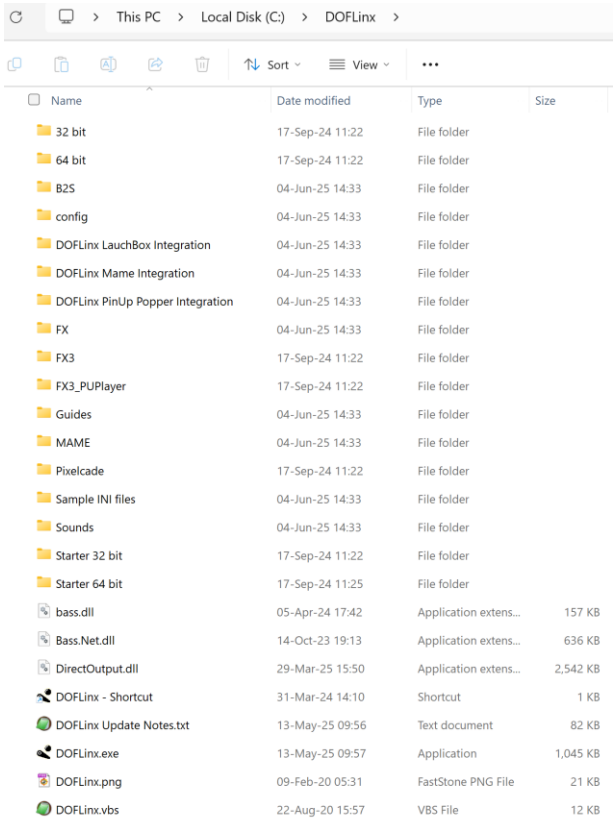
- **B2S Server** already installed https://github.com/vpinball/b2s-backglass/releases
- **DirectOutput** configured (have a look at **VirtuaTilt-RP2040_SetupGuide_v1.x.pdf**
- guide)
- **VTConfig-FX-FX3-DOFLinx_x1.X.json** profile loaded in **Pinscape Pico Config Tool**



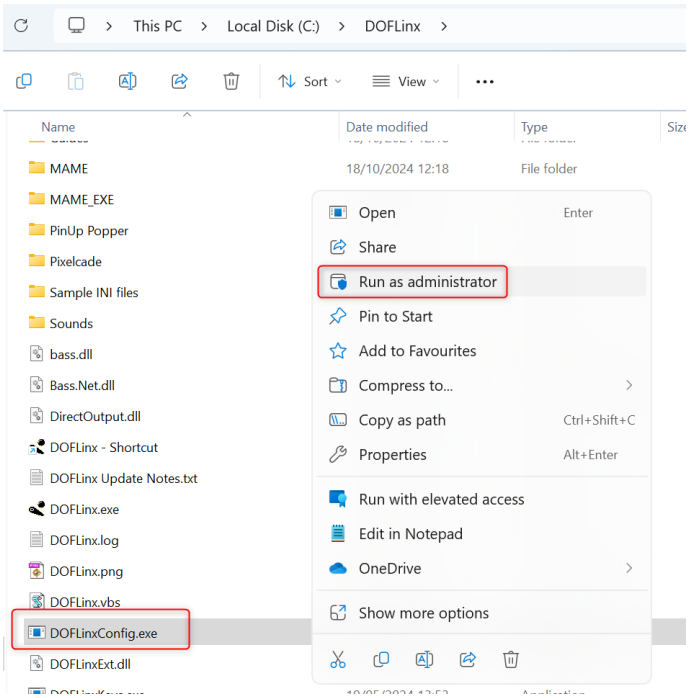

Import file
Local Disk (C:) › PinscapePico › + PROFILES
Search + PROFILES
Organize
New folder
Name
Date modified
Type
Size
VTConfig-FuturePinball-DOF_v1.1.json 22-Nov-25 03:09 JSON File 6 KB
VTConfig-FuturePinball-NODOF_v1.1.json 22-Nov-25 03:10 JSON File 6 KB
VTConfig-FX-FX3-DOFLinx_v1.1.json 22-Nov-25 03:10 JSON File 8 KB
VTConfig-FXVR_v1.0.json 11-Nov-25 10:35 JSON File 5 KB
VTConfig-IOS-Android_v1.2.json 19-Dec-25 10:47 JSON File 6 KB
VTConfig-PS_v1.2.json 05-Dec-25 03:41 JSON File 6 KB
VTConfig-STEAM-XInput_v1.1.json 22-Nov-25 03:14 JSON File 6 KB
VTConfig-Switch_v1.2.json 05-Dec-25 03:40 JSON File 6 KB
VTConfig-VPX-10.8.X-DOF_v1.1.json 22-Nov-25 03:16 JSON File 6 KB
VTConfig-VPX-NODOF_v1.1.json 22-Nov-25 03:17 JSON File 6 KB
VTConfig-VRClassics-StarWarsVR_v1.2.json 19-Dec-25 10:44 JSON File 6 KB
VTConfig-Xbox_v1.0.json 22-Nov-25 03:04 JSON File 6 KB
VTConfig-ZaccariaPinball_v1.1.json 22-Nov-25 03:19 JSON File 6 KB
File name:
JSON files (*.json)
Open
Cancel

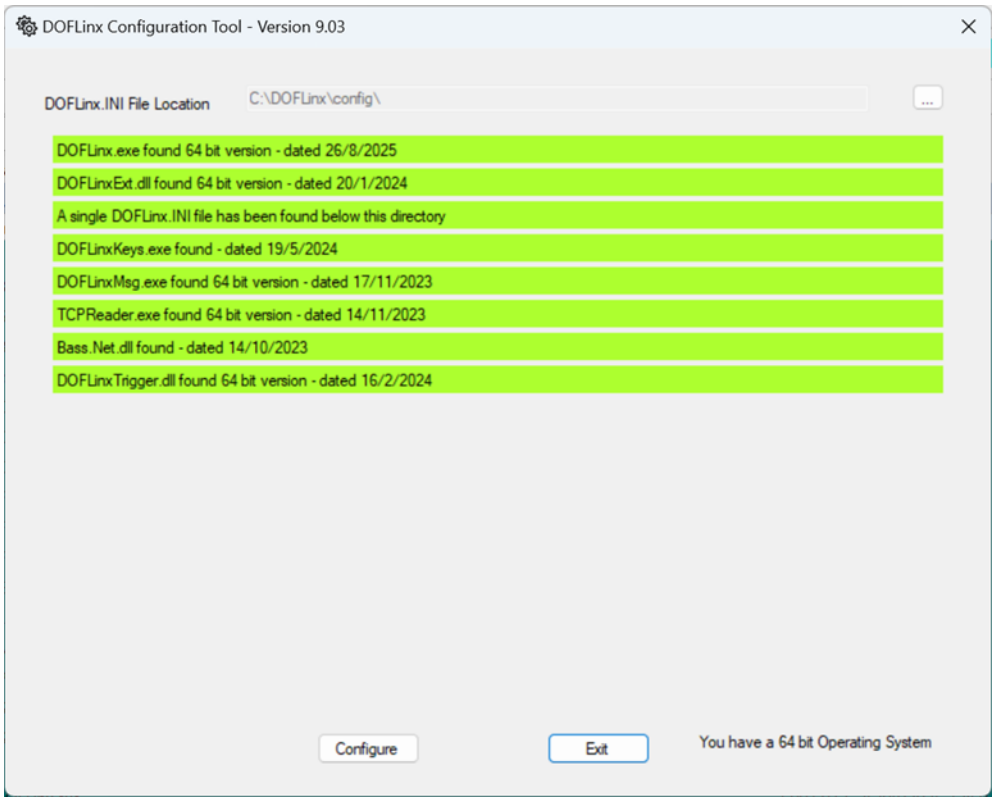

Pinscape Pico Config Tool
File Edit Device Backup Tools Help
Pinscape Pico
Unit #1
(Main Pico)
HW ID E66098F29B6C9C39
New Device Setup
Click to set up a new Pico
Overview Log Configuration* Safe-mode Config Buttons Nudge Output Ports Output Hardware Plunger
Program Pico
Find...
// Pinscape Pico Configuration - Starter Template
// Copyright 2024, 2025 Michael J Roberts / BSD-3-Clause license / NO WARRANTY
//
// VTConfig-FX-FX3-DOFLinx.json
{
id: {
unitNum: 1,
ledWizUnitNum: 1,
},
serialPorts: {
usb: {
logging: true,
console: {
Save to Pico
This will save the file to the Pico's flash memory and reboot the Pico to put the new settings into effect.
Do you wish to proceed?
Show this confirmation message before saving
Save to Pico
Cancel

Create a folder **C:\DOFLinx** and uncompress the above zip package (remember to unblock it):

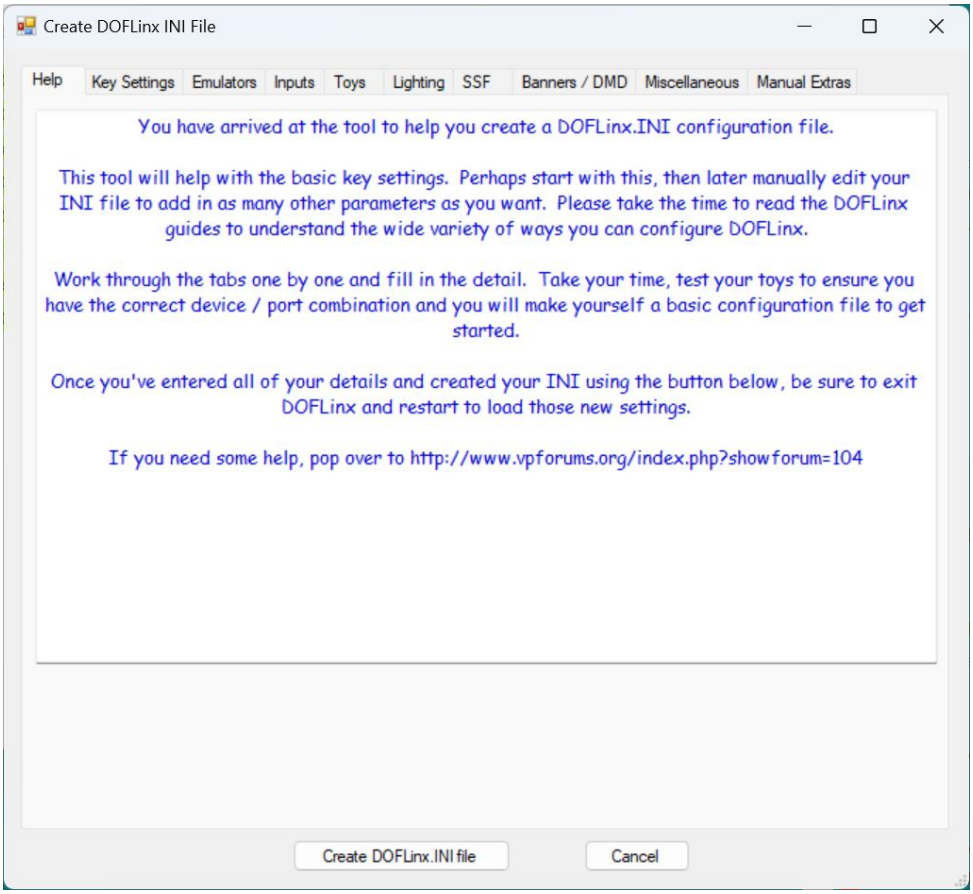


Run **DOFLinxConfig.exe** file in **DOFLinx** folder as administrator:

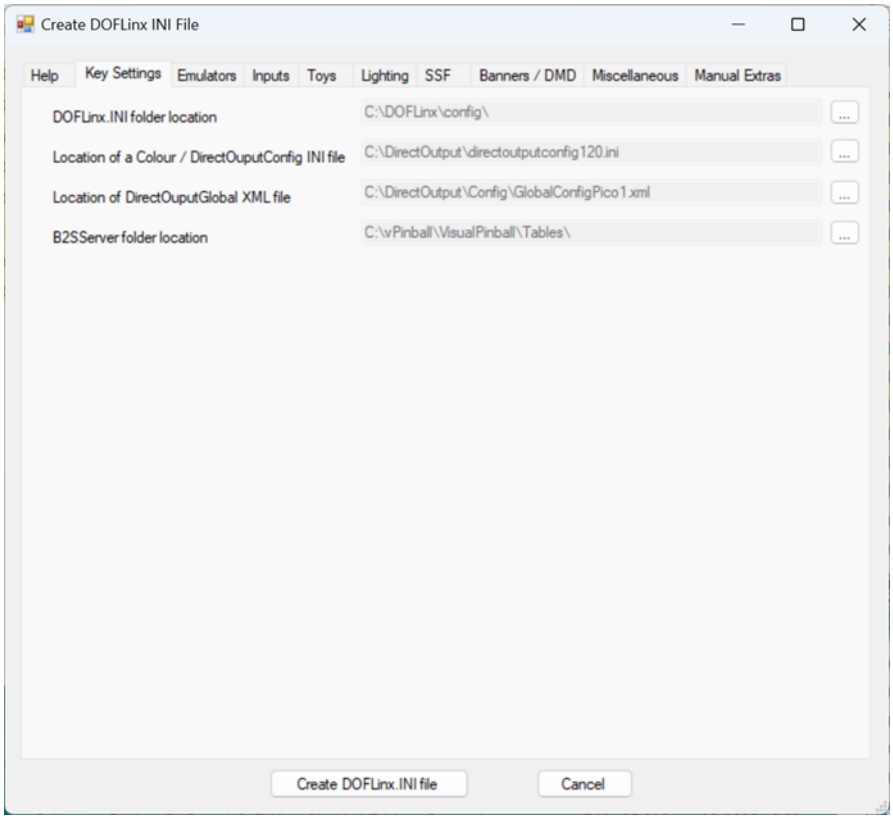


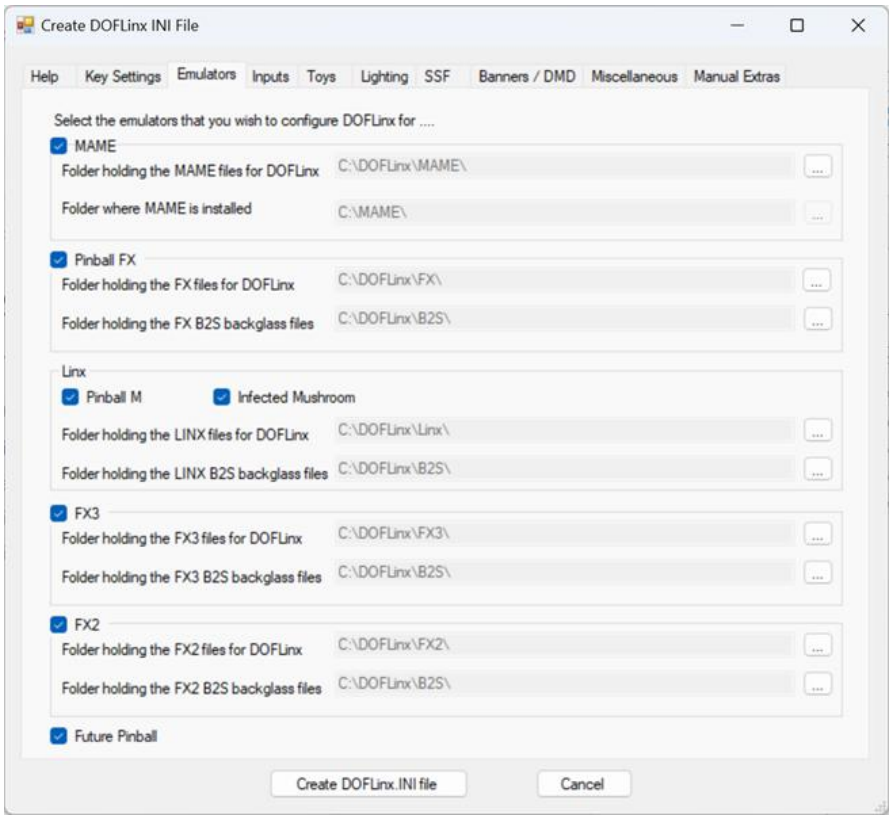


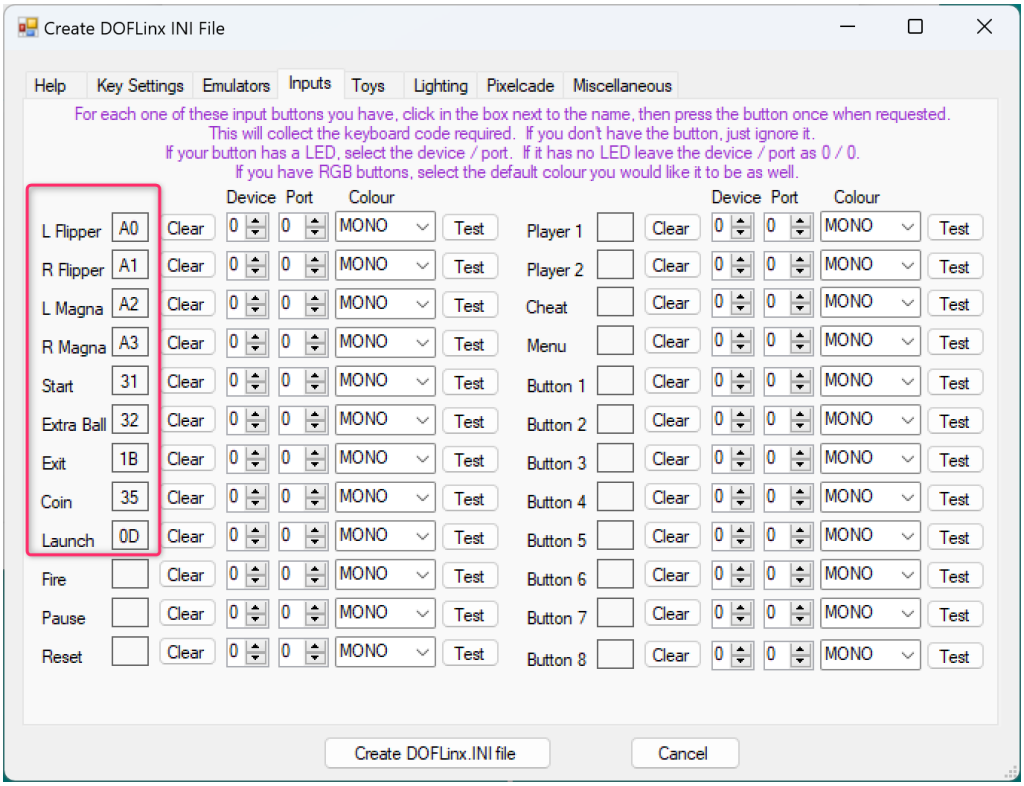
Turn on your VirtuaTilt power switch and click on **Configure button and work through all settings:**

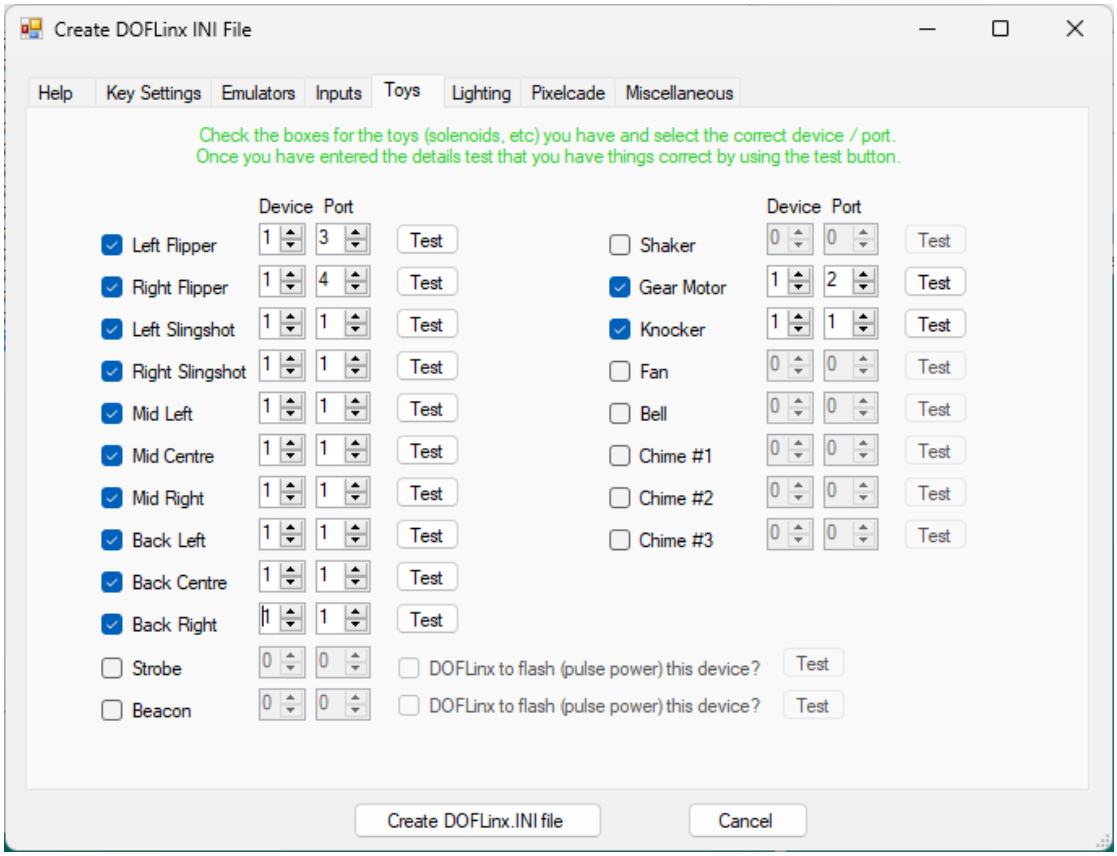


You need to configure **Key Settings**, **Emulators**, **Inputs** and **Toys** tabs:

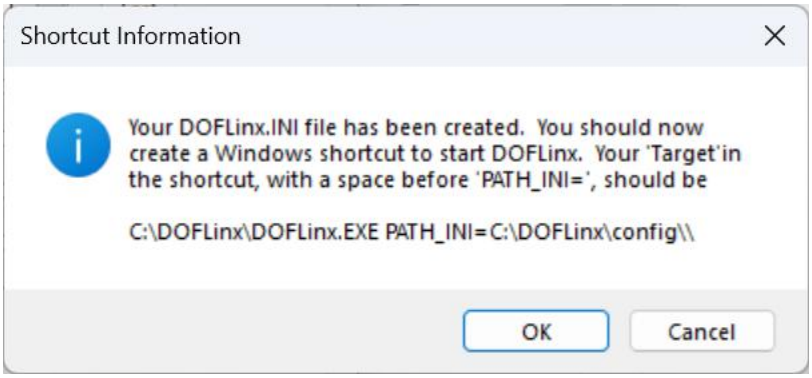



Create DOFLinx INI File
Help
Key Settings
Emulators
Inputs
Toys
Lighting
SSF
Banners / DMD
Miscellaneous
Manual Extras
Select the emulators that you wish to configure DOFLinx for ....
MAME
Folder holding the MAME files for DOFLinx
C:\DOFLinx\MAME\
Folder where MAME is installed
C:\MAME\
Pinball FX
Folder holding the FX files for DOFLinx
C:\DOFLinx\FX\
Folder holding the FX B2S backglass files
C:\DOFLinx\B2S\
Linx
Pinball M
Infected Mushroom
Folder holding the LINX files for DOFLinx
C:\DOFLinx\Linx\
Folder holding the LINX B2S backglass files
C:\DOFLinx\B2S\
FX3
Folder holding the FX3 files for DOFLinx
C:\DOFLinx\FX3\
Folder holding the FX3 B2S backglass files
C:\DOFLinx\B2S\
FX2
Folder holding the FX2 files for DOFLinx
C:\DOFLinx\FX2\
Folder holding the FX2 B2S backglass files
C:\DOFLinx\B2S\
Future Pinball
Create DOFLinx.INI file
Cancel

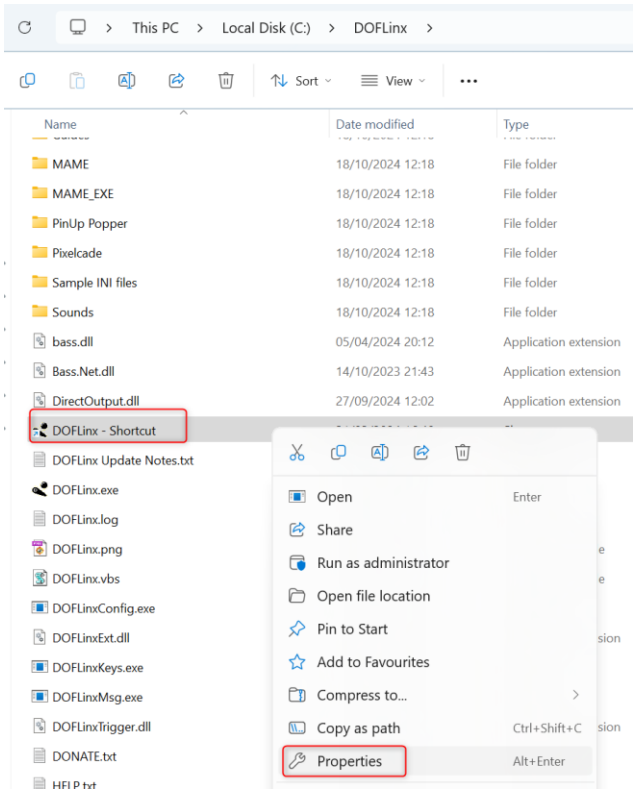

Create DOFLinx INI File
Help
Key Settings
Emulators
Inputs
Toys
Lighting
Pixelcade
Miscellaneous
For each one of these input buttons you have, click in the box next to the name, then press the button once when requested.
This will collect the keyboard code required. If you don't have the button, just ignore it.
If your button has a LED, select the device / port. If it has no LED leave the device / port as 0 / 0.
If you have RGB buttons, select the default colour you would like it to be as well.
Device Port Colour
L Flipper A0
R Flipper A1
L Magna A2
R Magna A3
Start 31
Extra Ball 32
Exit 1B
Coin 35
Launch 0D
Fire
Pause
Reset
Player 1
Player 2
Cheat
Menu
Button 1
Button 2
Button 3
Button 4
Button 5
Button 6
Button 7
Button 8
Clear
Test
MONO
Create DOFLinx.INI file
Cancel

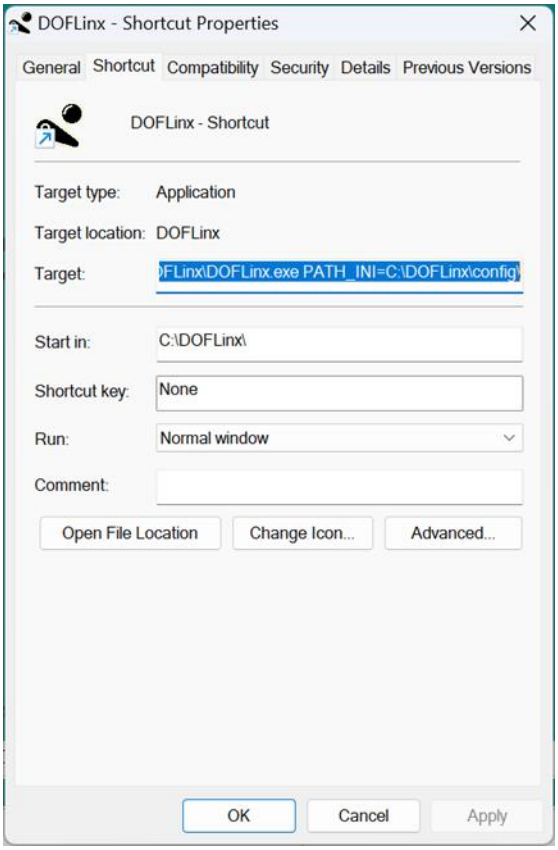


Click on "**Create DOFLinx .INI File**" when you're done:

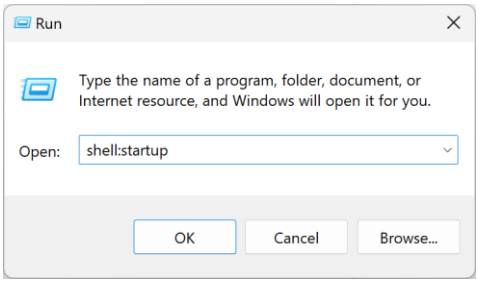


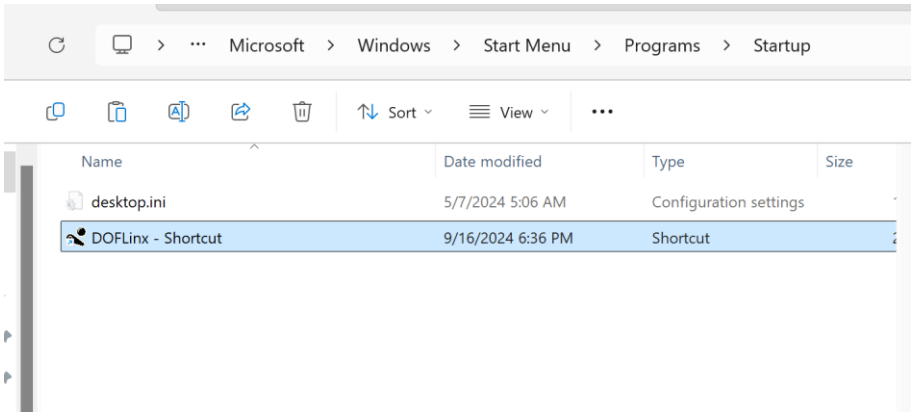
Edit properties of **DOFLinx – Shortcut.lnk** file and add the above string in Target field:



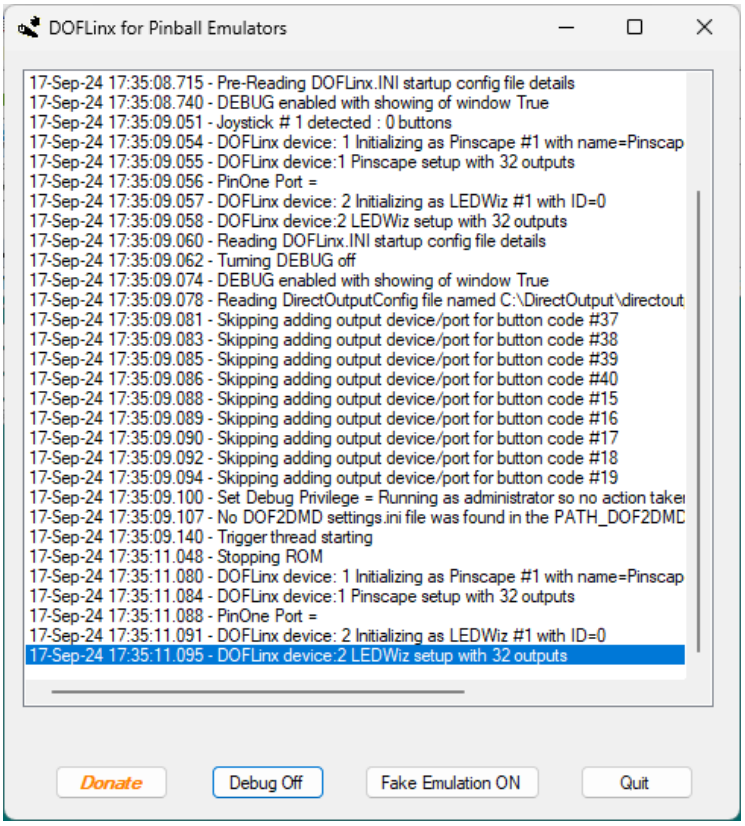


For automatic startup at every boot, put the shortcut in Startup folder:





Reboot the machine. If your DOFLinx.ini is configured correctly you should not have any error in debug logs (launch DOFLinx - Shortcut to check):



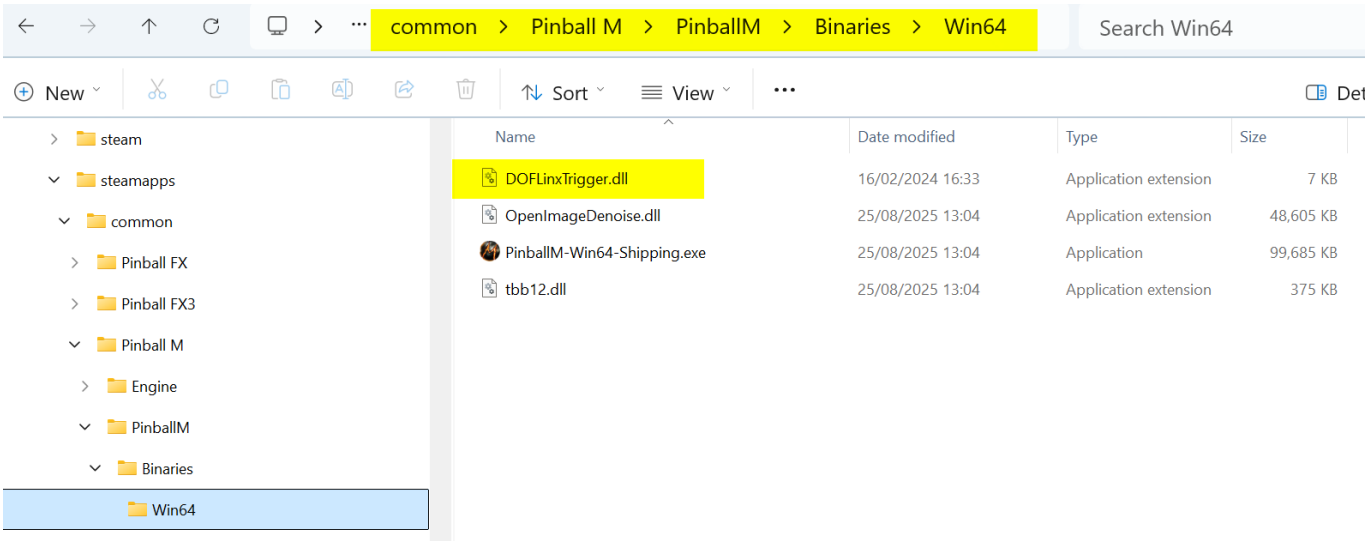
Accelerometer/Analog Plunger configuration

Pinball M emulator do not support analog plunger natively. However, new VirtuaTilt has native support for accelerometer and plunger (**X360CE** emulator is not needed anymore)!

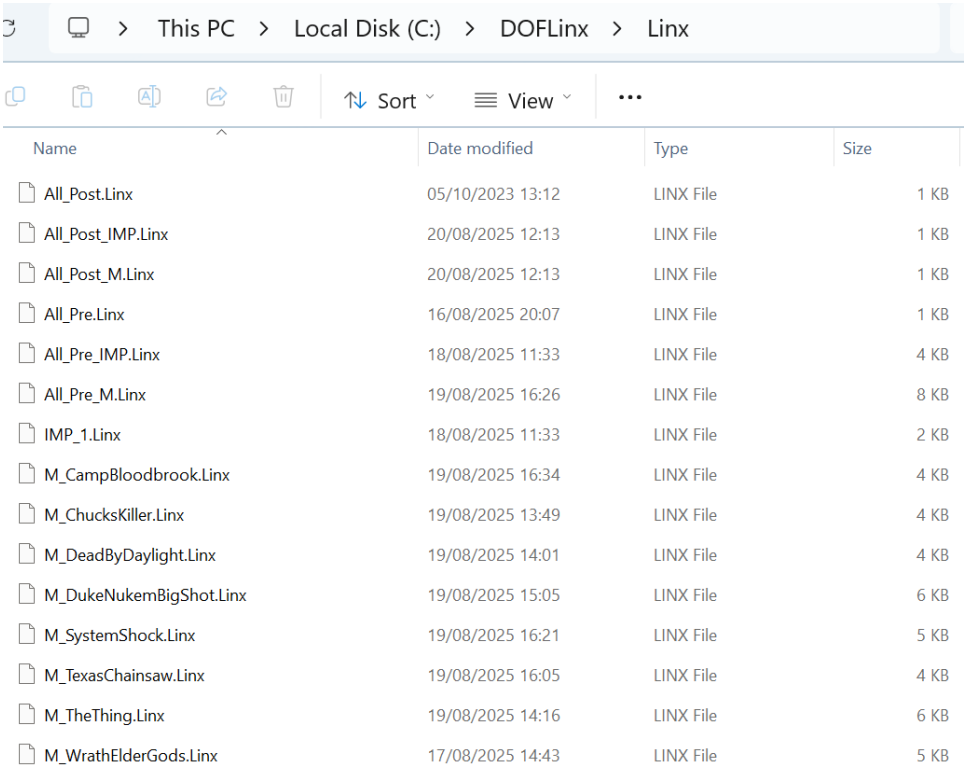
Enabling Pinball M for DOFLinx

Copy **DOFLinxTrigger.dll** from where you unzipped the DOFLinx download to the same folder

as PinBallM-Win64-Shipping.exe **(C:\Program Files (x86)\Steam\steamapps\common\Pinball M\PinballM\Binaries\Win64\)**



Make sure that you've got the LINX files from the \LINX folder in the DOFLinx download zip in a folder where you want them, ie **C:\DOFLinx\Linx**



Edit your **DOFLinx.INI** to make Pinball M activate DOFLinx by adding in the **PROCESSES=** line **PinballM-Win64-Shipping**, so if you want for example MAME and PinballM to activate DOFLinx you would have:

PROCESSES=Mame,PinballM-Win64-Shipping

Note: **PinballM-Win64-Shipping** is the process to monitor for, but you still run **PinballM.exe** to start PinballFX.

Be sure to have the **PATH_LINX=** and **PATH_LINX_B2S=** parameters pointing to the place you put the Pinball M files.

Make sure you have a **** on the end of the path, so **C:\DOFLinx\Linx**

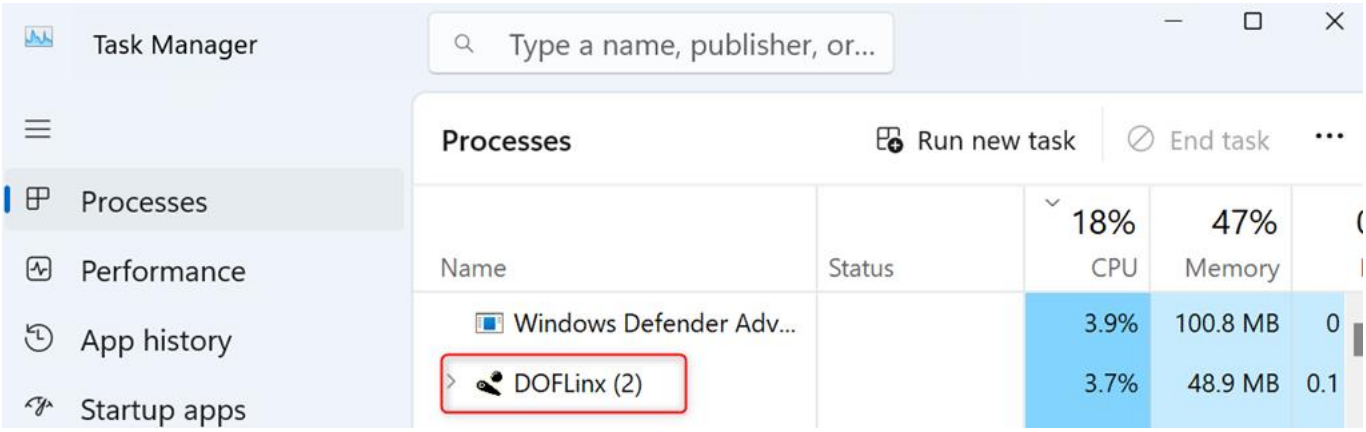
```
PATH_PIXELCADE=C:\Pixelcade\
PATH_HI2TXT=C:\HI2TXT\
PATH_LINX=C:\DOFLinx\Linx\
PATH_LINX_B2S=C:\DOFLinx\B2S\

MAME_PROCESS=Mame
PROCESSES=PinballFX-Win64-Shipping,Pinball FX3,Pinball FX2,Future Pinball
PROCESSES_LINX=PinballM-Win64-Shipping,IMPinball-Win64-Shipping
```

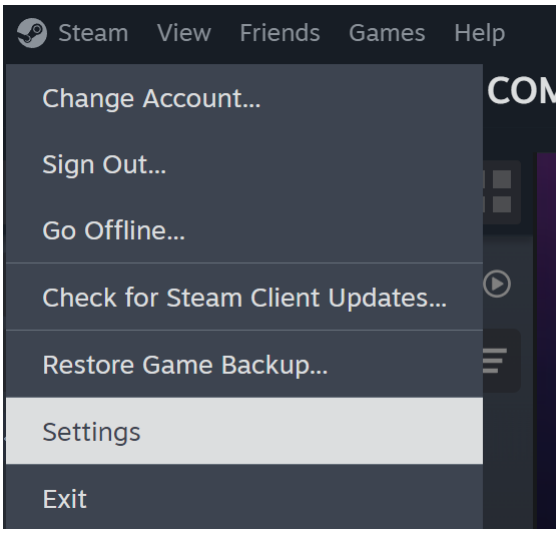
You can get the complete set of active backglasses for Pinball M from

https://github.com/DOFLinx/B2S-Back-Glasses/releases and place them in the folder you just set up above.

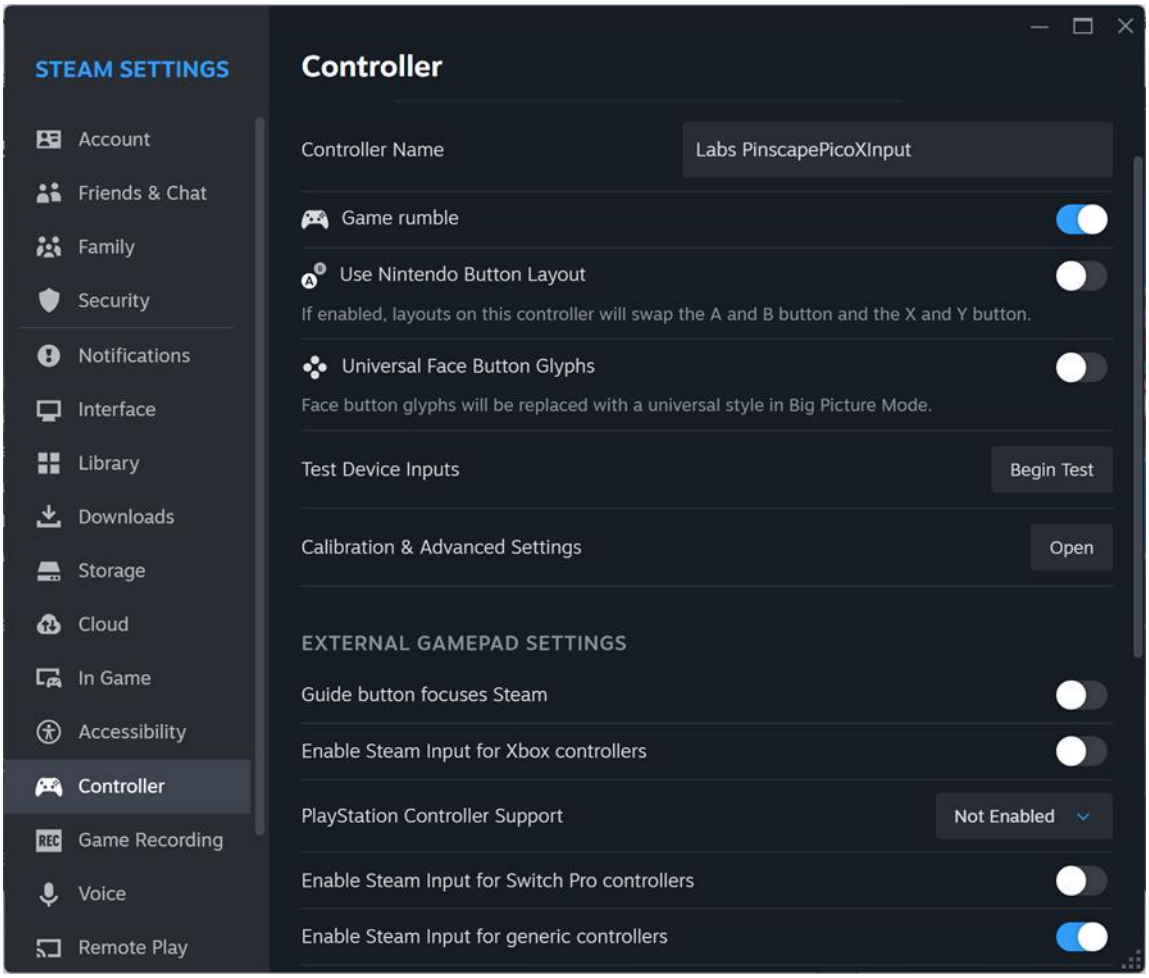
Reboot the machine and check that **DOFLinx process** is running:



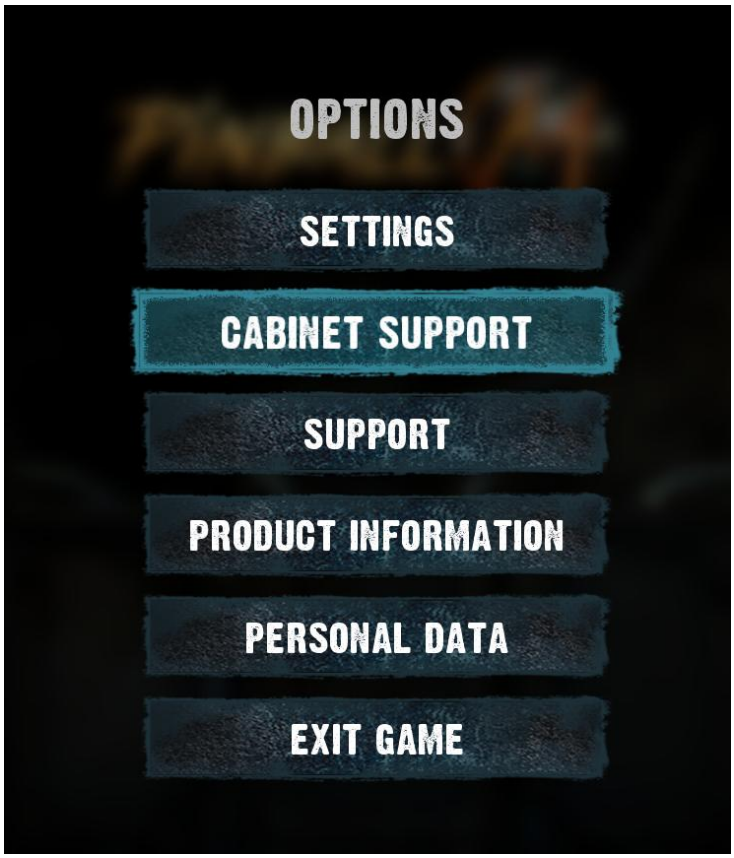
Go to Steam **Settings → Controller**

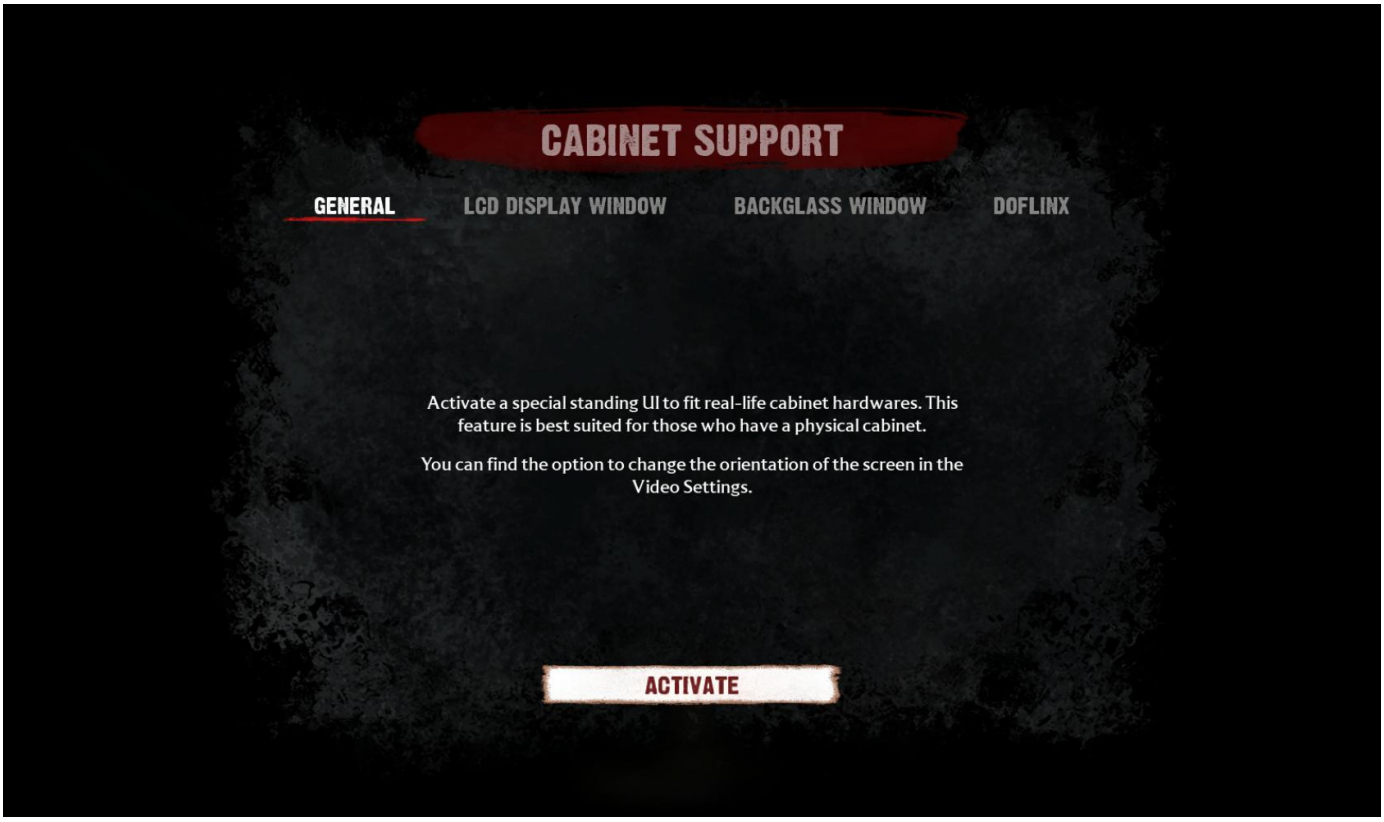


Disable everything under "External Gamepad Settings" with the except of "**Enable Steam Input for generic controllers**".

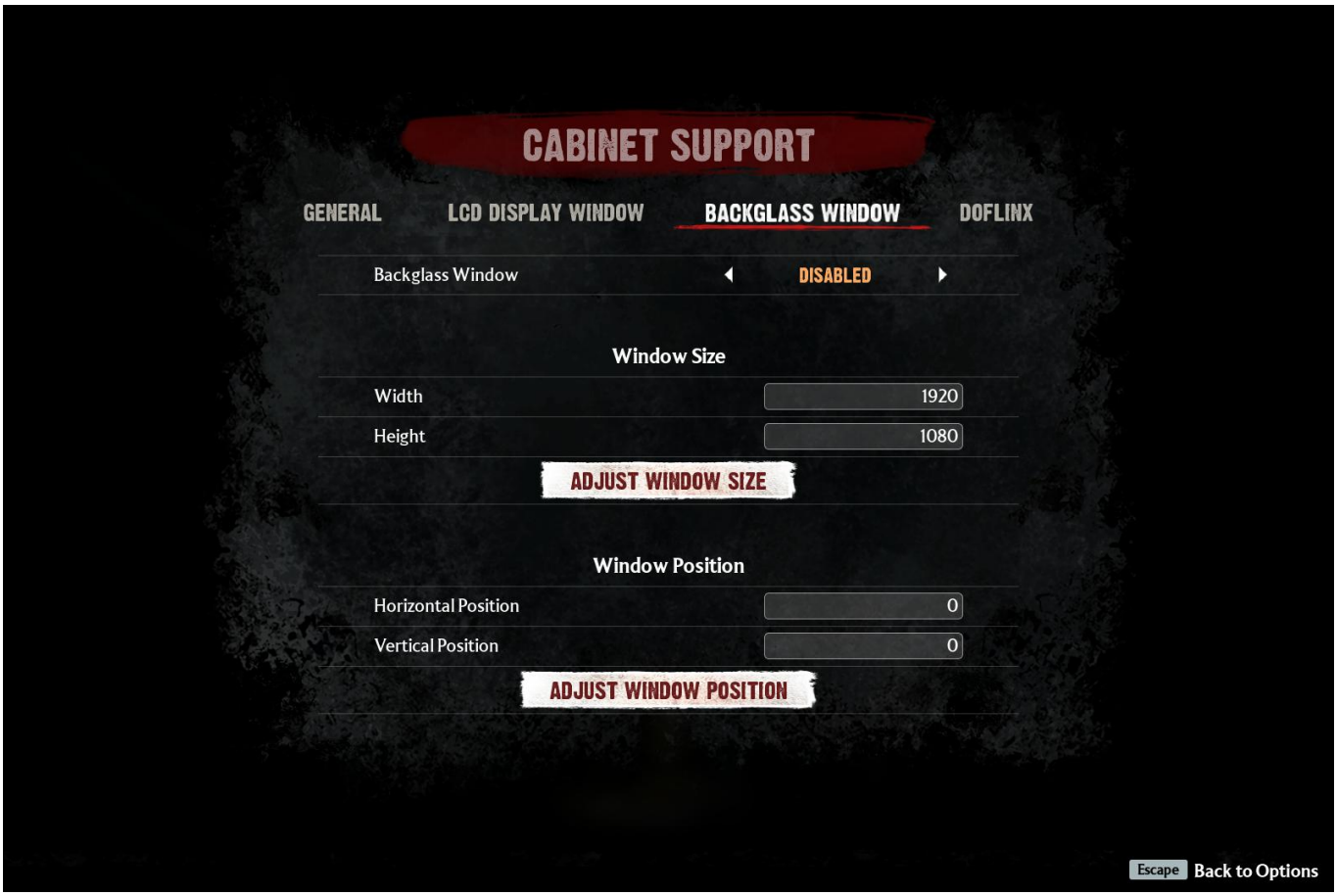


Start Pinball M, go to **Options | Cabinet Support | General** and **activate cabinet mode:**

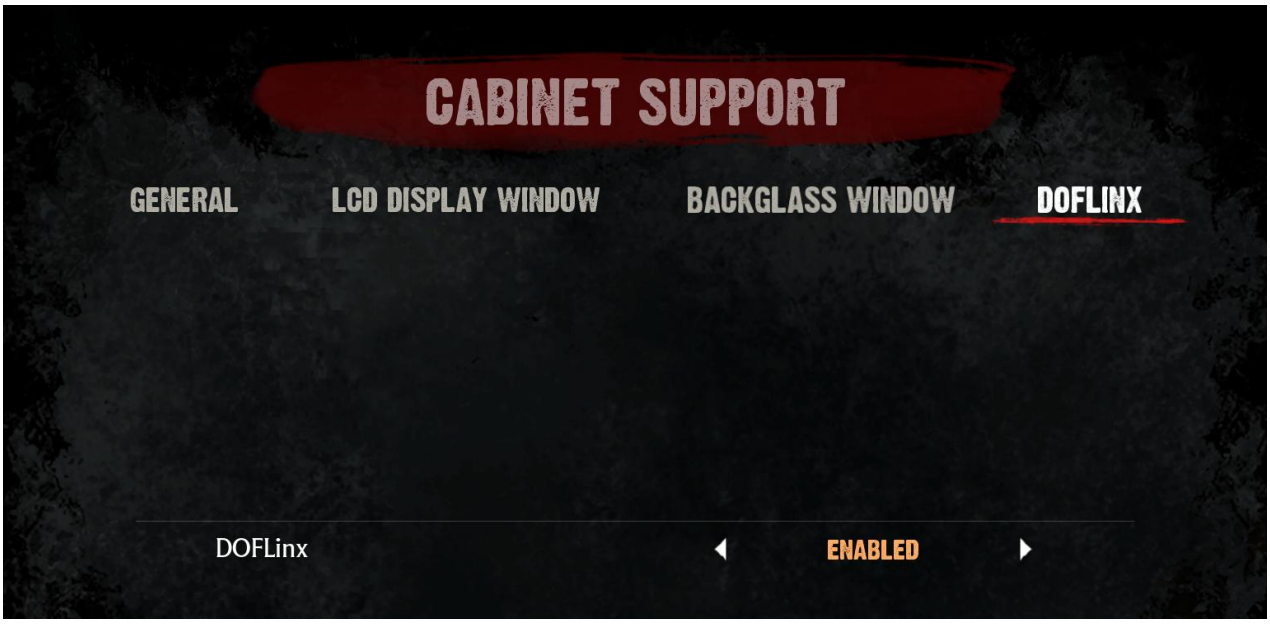




Disable the Backglass Window as you will use the DOFLinx active backglasses:



Go to the **DOFLinx** tab, **enable it**:



Start Pinball M and have fun with full DOFLinx feedback!

If everything was setup correctly you should see DOFLinx logo while loading:



Troubleshooting Tips

If DOFLinx logo is present but DOFLinx feedback is not working, quit the game then kill DOFLinx process in Task Manager and run it again manually from C:\DOFLinx folder. Then restart the game.

DOFLinx.INI file for VirtuaTilt

You can have a look at the following **DOFLinx.ini** file as a reference for your VirtuaTilt. Be sure to correctly configure all paths:

##

#

The config file for DOFLinx Starter MAME, Pinball FX, Pinball M and Infected Mushroom Pinball

To find out more about each parameter have a read of the included main guide 'DOFLinx Guide'

#

##

0 = OFF (normal operation), 1 = ON - When ON a window will appear and a log file will be created in the EXE directory.

DEBUG=0

#location of some general files

COLOUR_FILE=C:\DirectOutput\directoutputconfig120.ini

DIRECTOUTPUTGLOBAL=C:\DirectOutput\Config\GlobalconfigPico1.xml

PATH_B2S_SERVER=C:\vPinball\VisualPinball\Tables

#location of emulators files

PATH_FX2=C:\DOFLinx\FX2

PATH_FX2_B2S=C:\DOFLinx\B2S

PATH_FX3=C:\DOFLinx\FX3

PATH_FX3_B2S=C:\DOFLinx\B2S

PATH_MAME=C:\DOFLinx\MAME

FP_ATTEMPT_LINK=1

MAME_FOLDER=C:\MAME

PATH_PIXELCADE=C:\Pixelcade

PATH_HI2TXT=C:\HI2TXT

PATH_LINX=C:\DOFLinx\Linx

PATH_LINX_B2S=C:\DOFLinx\B2S

MAME_PROCESS=Mame

PROCESSES=PinballFX-Win64-Shipping,Pinball FX3,Pinball FX2,Future Pinball

PROCESSES_LINX=PinballM-Win64-Shipping,Pinball-Win64-Shipping

#If you want the MAME score screen to appear then set the below to 1

SCORE_SCREEN=0

L_FLIPPER_KEY=A0

R_FLIPPER_KEY=A1

LINK_BUT_LF=0000,MONO,A0

LINK_BUT_RF=0000,MONO,A1

LINK_BUT_ML=0000,MONO,A2

LINK_BUT_MR=0000,MONO,A3

LINK_BUT_ST=0000,MONO,31

LINK_BUT_EB=0000,MONO,32

LINK_BUT_EX=0000,MONO,1B

LINK_BUT_CN=0000,MONO,35

LINK_BUT_LB=0000,MONO,0D

L_FLIPPER_OUTPUT=1003

R_FLIPPER_OUTPUT=1004

LINK_LF=1003,100,10000,255

LINK_RF=1004,100,10000,255

LINK_LS=1001,100,500,255

LINK_RS=1001,100,500,255

LINK_ML=1001,100,500,255

LINK_MC=1001,100,500,255

LINK_MR=1001,100,500,255

LINK_BL=1001,100,500,255

LINK_BC=1001,100,500,255

LINK_BR=1001,100,500,255

LINK_GR=1002,1000,10000,255

LINK_KN=1001,100,500,255

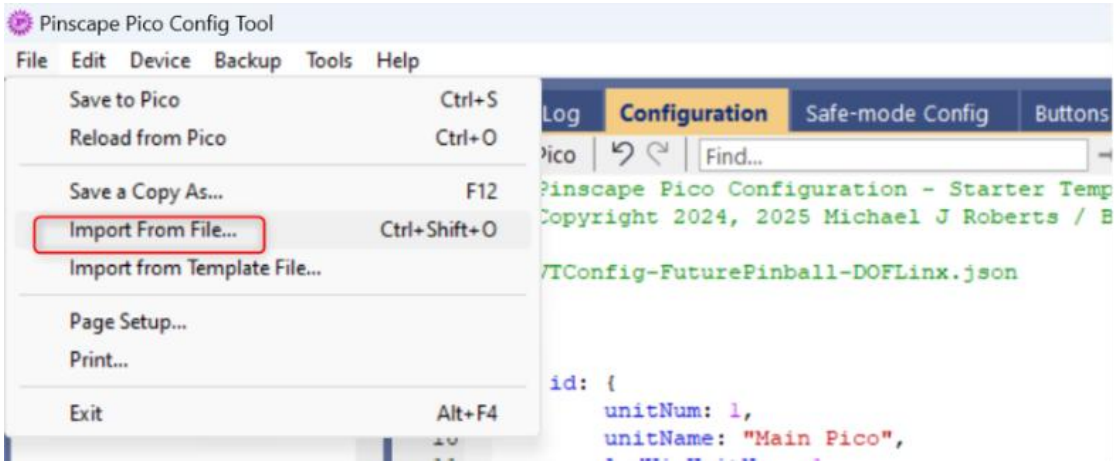
#!!!!!!!!!!!!!!!!!!!!!!!!!!!!

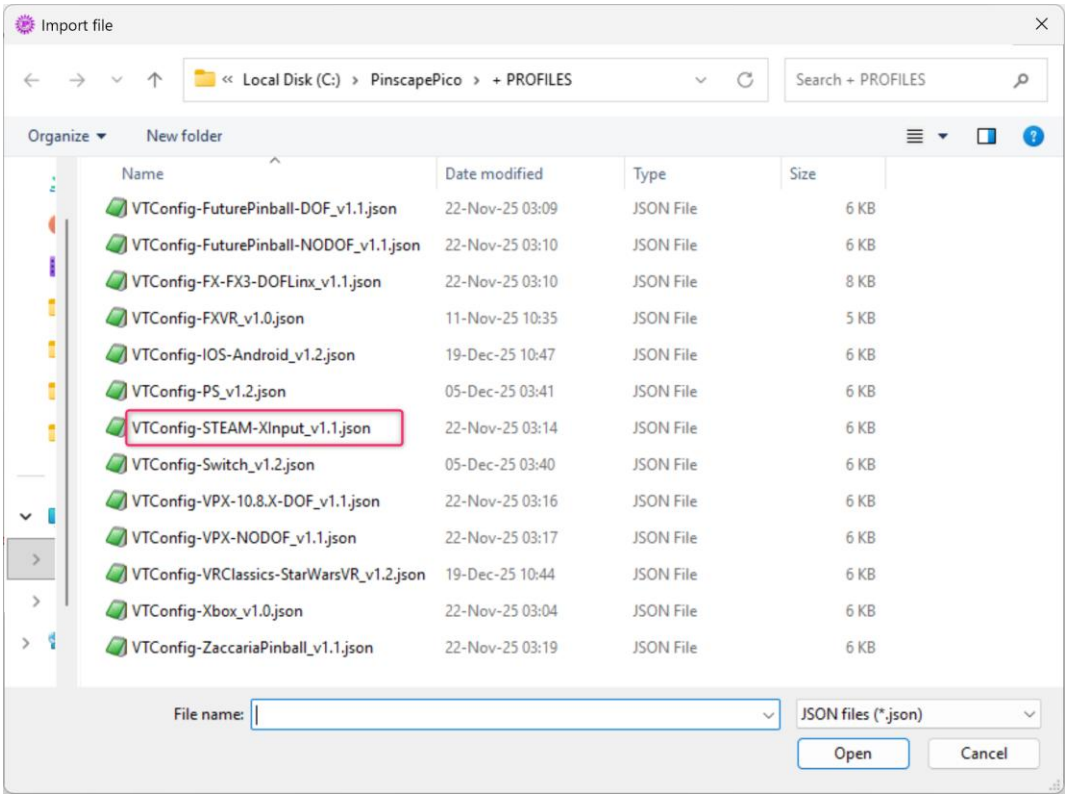
Use VirtuaTilt without DOF/DOFLinx

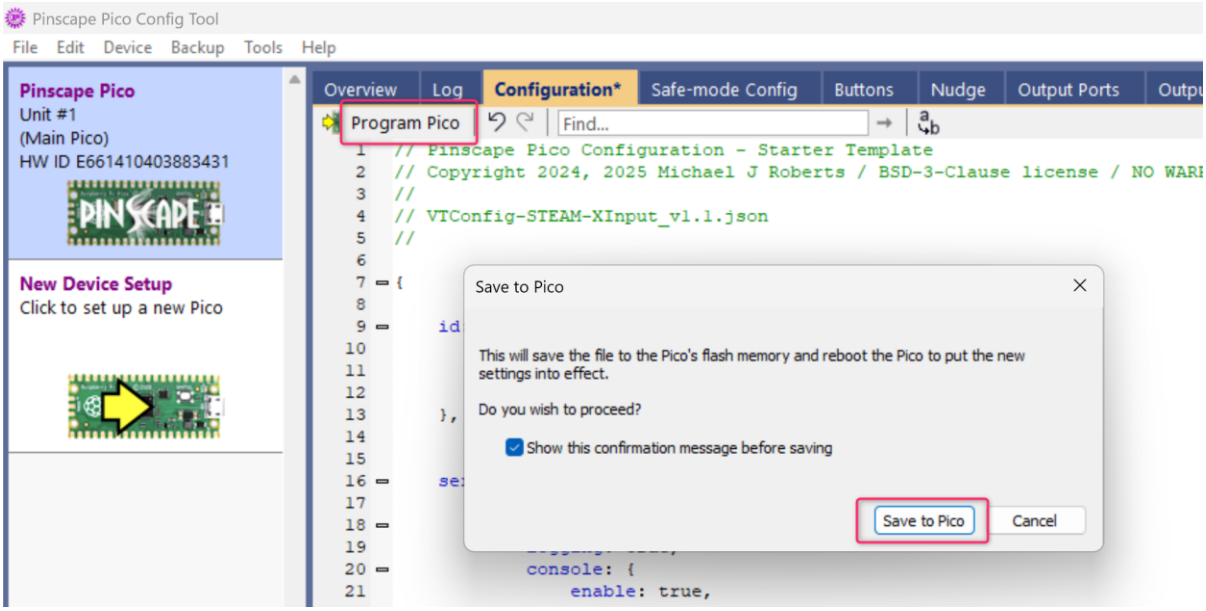
One of the main features of the new VirtuaTilt is the possibility to use it without going into DOF/DOFLinx configurations and benefit of feedback from solenoids and shaker motor, as well as analog plunger and accelerometer nudging. Just Plug&Play !

The **XInput mode** is used in Pinscape Pico firmware to achieve that.

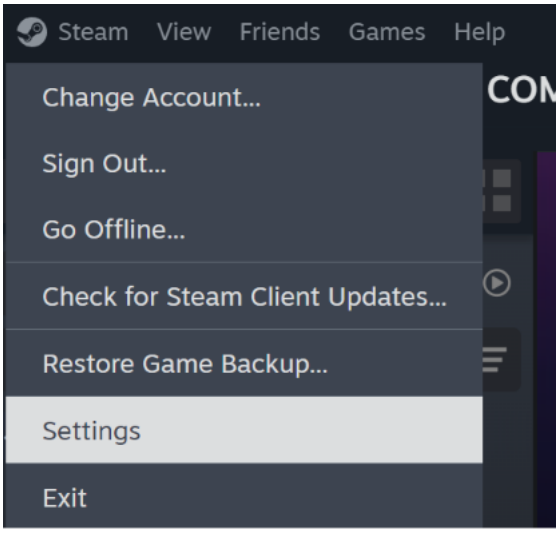
What you need to do is load **VTConfig-STEAM-XInput_v1.X.json** profile in **Pinscape Pico Config Tool** :



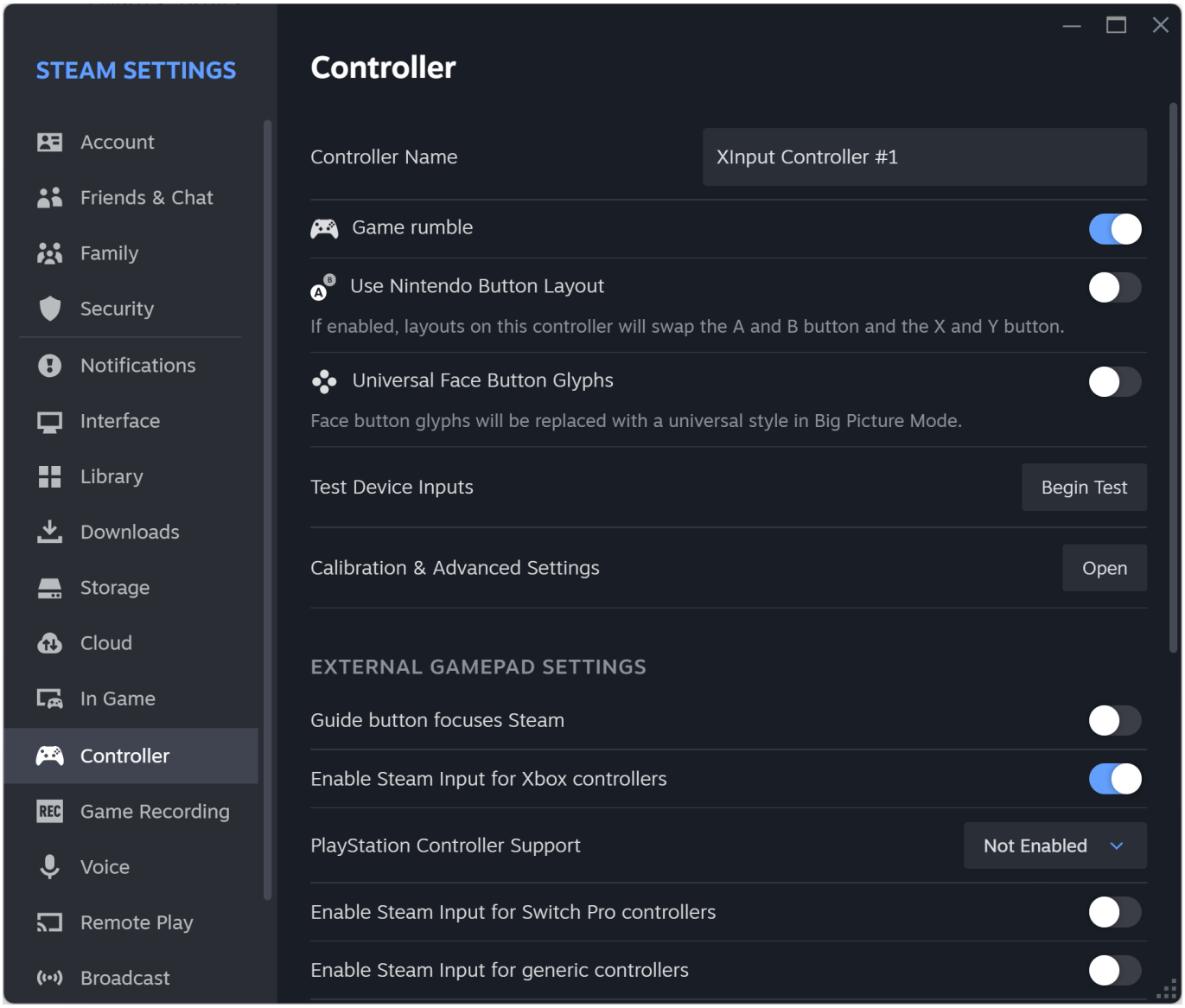




Start **Steam** and go to **Settings → Controller**



Disable everything under "External Gamepad Settings" with the except of "**Enable Steam Input for Xbox controllers**" and "**Game rumble**":



That's it! Launch your favorite Virtual Pinball game and have fun.

IMPORTANT NOTE: **Xbox drivers for Windows installation is required** (see **Xbox Drivers Installation** chapter in Main Setup Guide).